Solution: Pangram

Problem Statement

Given a string `sentence` containing English letters (lower- or upper-case), return `true` if `sentence` is a **Pangram**, or `false` otherwise.

A **Pangram** is a sentence where every letter of the English alphabet appears at least once.

Note: The given `sentence` might contain other characters like digits or spaces, your solution should handle these too.

Example 1:

```
Input: sentence = "TheQuickBrownFoxJumpsOverTheLazyDog"
Output: true
Explanation: The sentence contains at least one occurrence of every
letter of the English alphabet either in lower or upper case.
```

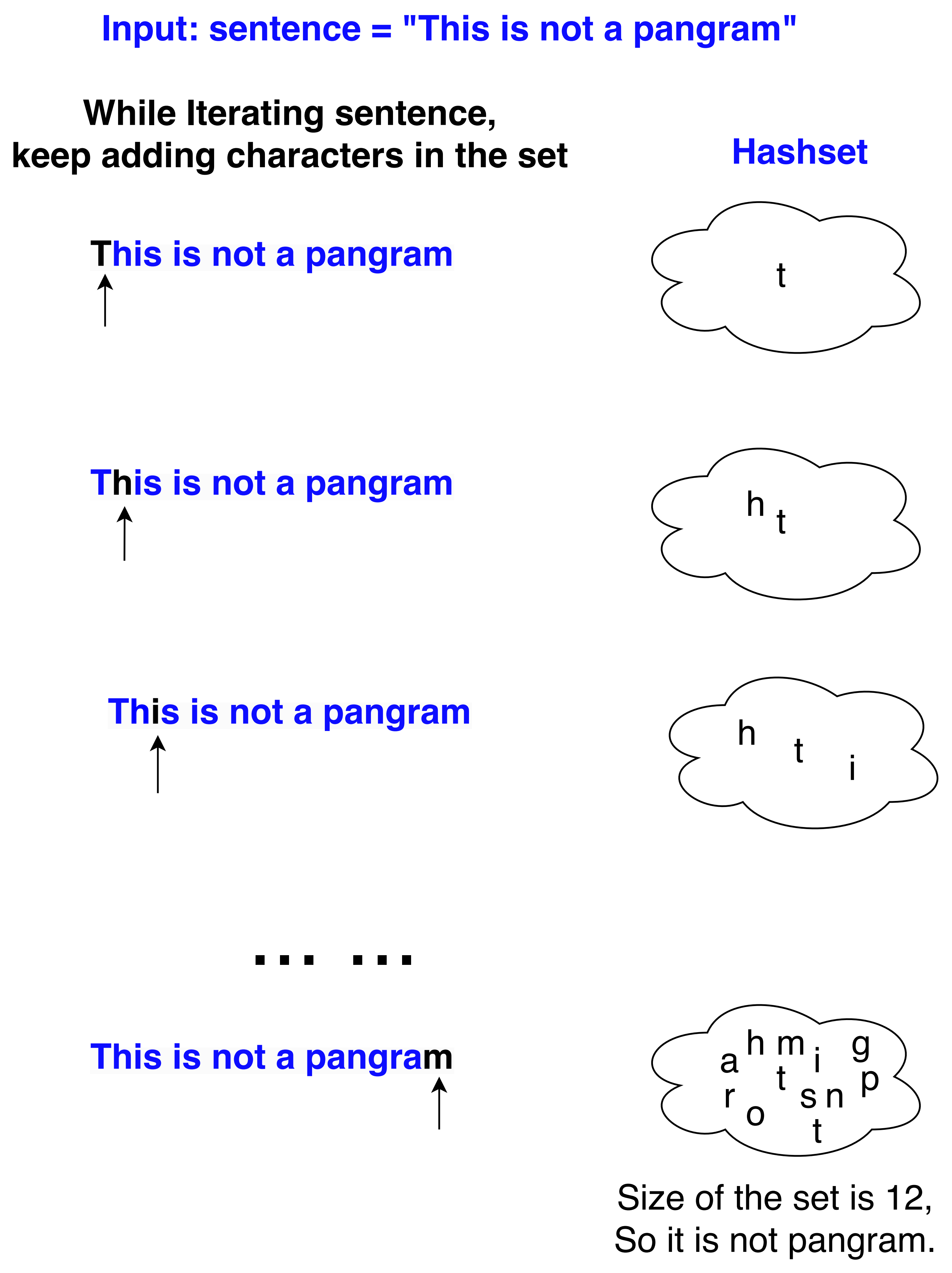
Example 2:

```
Input: sentence = "This is not a pangram"
Output: false
Explanation: The sentence doesn't contain at least one occurrence of
every letter of the English alphabet.
```

Solution

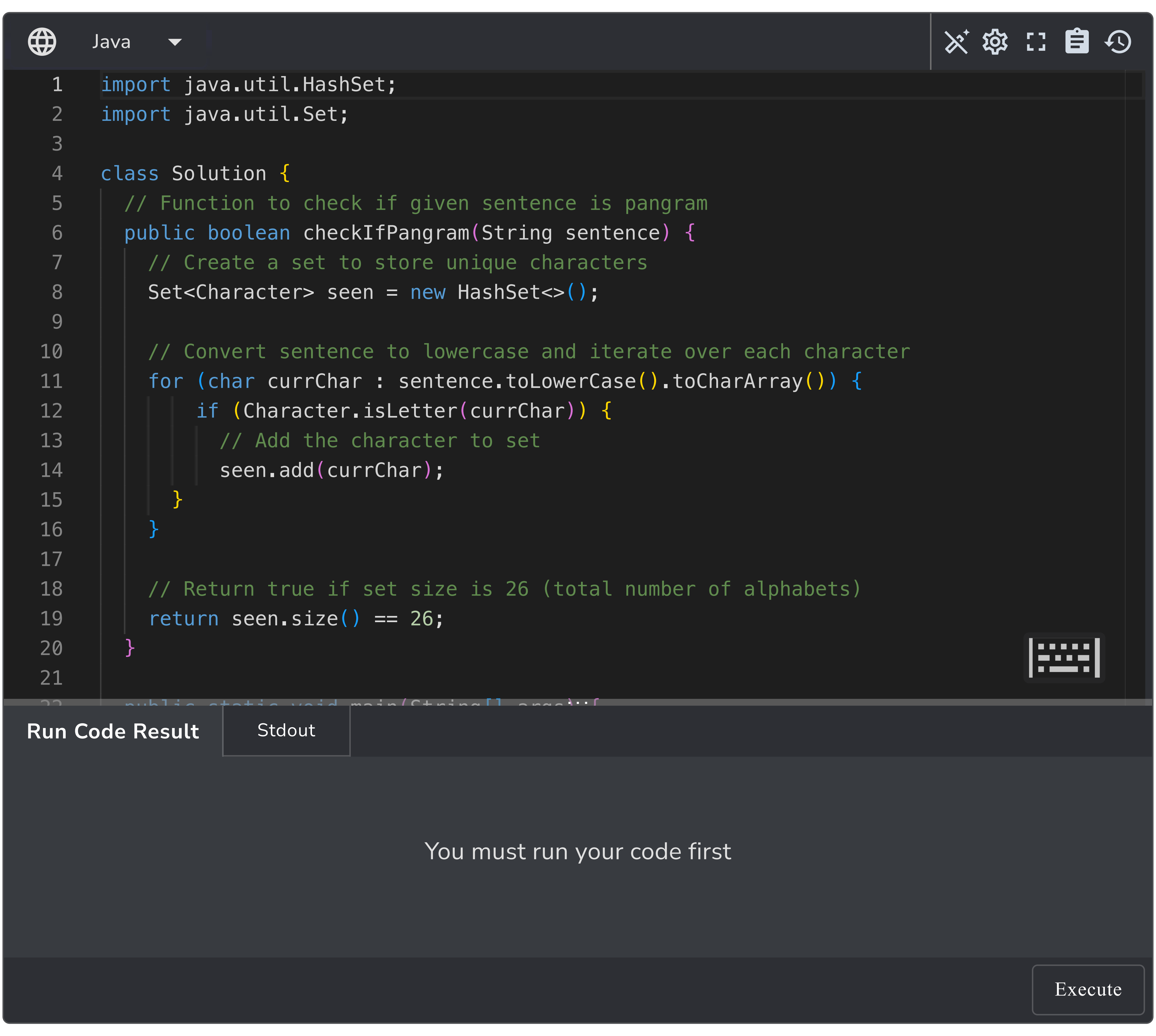
We can use a `HashSet` to check if the given sentence is a pangram or not. The `HashSet` will be used to store all the unique characters in the sentence. The algorithm works as follows:

1. Converts the sentence to lowercase.
2. Iterate over each character of the sentence using a loop.
3. Add each character to the `HashSet`.
4. After looping through all characters, compare the size of the `HashSet` with 26 (total number of alphabets). If the size of the HashSet is equal to 26, it means the sentence contains all the alphabets and is a pangram, so the function will return true. Otherwise, it will return false.



Code

Here is the code for this algorithm:



```java
import java.util.HashSet;
import java.util.Set;

class Solution {
  // Function to check if given sentence is pangram
  public boolean checkIfPangram(String sentence) {
    // Create a set to store unique characters
    Set<Character> seen = new HashSet<>();

    // Convert sentence to lowercase and iterate over each character
    for (char currChar : sentence.toLowerCase().toCharArray()) {
        if (Character.isLetter(currChar)) {
          // Add the character to set
          seen.add(currChar);
        }
    }

    // Return true if set size is 26 (total number of alphabets)
    return seen.size() == 26;
  }

```

Time Complexity

1. **Iterating Over Characters:** The main operation in the code is iterating over each character in the input string. If the length of the input string is `n`, this iteration occurs `n` times.

2. **Set Operations:** For each character, the code performs a constant-time operation—adding the character to a HashSet if it's a letter. The time complexity for adding an element to a HashSet is typically $O(1)$.

3. **Overall Time Complexity:** Considering the iteration over `n` characters and constant-time set operations, the total time complexity is $O(n)$, where `n` is the length of the sentence.

Space Complexity

1. **HashSet Storage:** The HashSet `seen` is used to store the distinct characters encountered in the sentence. In the worst-case scenario, it will store all `26` letters of the alphabet.

2. **Constant Size Set:** Regardless of the input sentence length, the HashSet can only grow up to a size of `26`. This is because it only stores distinct English alphabet letters.

3. **Overall Space Complexity:** Given the HashSet's maximum size is constant (at most `26` characters), the space complexity is $O(1)$, meaning it is constant.

Conclusion

- **Time Complexity:** $O(n)$, where `n` is the length of the input string.
- **Space Complexity:** $O(1)$ (constant space, independent of input string length).

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